

Reduced Cas9 transgene silencing by incorporation of intron sequences

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Silencing remains a significant challenge for exogenous gene expression, limiting both the penetrance and expressivity of transgenes. In particular, silencing of Cas9 expression is a major technical limitation for many gene editing and CRISPR screening applications. Here, we demonstrate that including introns in Cas9 expression cassettes significantly reduces silencing across multiple cell lines. Notably, the incorporation of an intron into a CRISPRa construct results in reduced silencing, increased expression levels, and markedly enhanced activation of target genes. We investigate diverse intron sequences and discover that T-rich introns over 2 kb confer the greatest protection against silencing. In addition, we find that introns can work synergistically with chromatin opening elements to further mitigate silencing, suggesting regulatory mechanisms are acting at both the DNA and RNA level to silence exogenous genes. Our work highlights the potential of introns to optimize genetic constructs for enhanced expression and improved cellular engineering requiring constitutive expression of large transgenes.

Cells have a myriad of machinery that silences the expression of exogenous genes. Some of these systems likely evolved from a need to guard the genome against the spread of mobile genetic elements or viral infections. However, these same mechanisms present a challenge for biotechnology applications, where robust transgene expression is desired. Transgene silencing can manifest in two forms: as a binary event, where cells harboring stably integrated transgenes stop expressing the corresponding protein, or as reduced expression of the transgene protein within individual cells^{1,2}. Both of these can cause problems in cell engineering technologies, such as the creation of synthetic gene circuits, biomanufacturing, and gene therapy³.

The mechanisms of gene silencing are diverse, ranging from chromatin remodeling to DNA and histone methylation to the spread of heterochromatin³. Although accessible chromatin is a requirement

for transcription, even transgenes integrated into accessible “safe harbor” sites are subject to silencing, suggesting active mechanisms of detection and repression of exogenous genes^{4,5}. While transgene silencing occurs through multiple mechanisms, the human silencing hub (HUSH) complex has been recently suggested to be responsible for recognition and repression of long, intronless transgenes^{6,7}. These single-exon, intronless transcripts that are the hallmark of reverse transcription are similar to the transgenes used in synthetic biology to express tools or proteins for genome engineering, and are therefore targeted by HUSH for silencing^{8,9}. HUSH recognizes foreign RNA transcripts, such as long interspersed element-1 (LINE1) retrotransposons, and recruits SETDB1 to silence the locus through methylation of histone H3 at lysine 9 (H3K9)⁶. HUSH preferentially silences transcripts more than 1 kb in length and sequences with high A

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nucleotide content⁷. While the mechanisms by which HUSH recognizes a genetic element as foreign are still under investigation, evidence suggests that a lack of introns might be a distinguishing feature of “non-self” RNA that allows HUSH to recognize these foreign transcripts^{6,7}.

A variety of evidence suggests that the presence of introns can help to increase transgene expression in different organisms, through a process known as intron-mediated enhancement¹⁰. Commonly used mammalian promoters EF1a, CAG, and UbC include introns and are used to drive high levels of expression across many cell types^{11,12}. Introns are used extensively in non-mammalian model organisms such as various plant species^{13–18}, *Drosophila*¹⁹, and *C. elegans*^{20,21} to enhance gene expression. Although it has been reported that introns also enhance expression in mammalian cells^{22,23}, the use of introns remains uncommon in bioengineering of transgenes for expression in mammalian cell lines, potentially because it has not been clear which introns function to enhance expression or where they should be placed in the cassette for optimal expression.

Cas9 transgenes are widely used in mammalian genome engineering, but expression of these constructs is frequently silenced in many cell types^{3,24,25}. This presents a major limitation for CRISPR screening, gene therapy, and synthetic biology applications. Here, we test whether Cas9 silencing could be mitigated by the inclusion of introns. We compare insertion of the second intron of the human β -globin gene (*HBB* IVS2) with a panel of other introns spanning different lengths and sequences into Cas9 at different positions and assess their ability to reduce silencing. We identify a number of effective introns which can significantly inhibit silencing, resulting in increased efficacy of nuclease Cas9, as well as more potent gene activation by CRISPRa. Together, these results demonstrate a simple modification to mammalian expression vectors that can greatly enhance transgene expression for diverse genome engineering applications.

Results

Addition of the *HBB* IVS2 intron reduces Cas9 silencing and does not interfere with Cas9 activity

To assess silencing of Cas9, we used the large serine recombinase Cp36²⁶ to randomly integrate a bicistronic cassette constitutively expressing either EF1a-driven Cas9-BFP (4.9 kb) or BFP alone (696 bp) and UbC-driven blasticidin S deaminase (BSD) (396 bp), which was used for antibiotic selection of successful integrations (Fig. 1a, b). Given the recent demonstration that HUSH preferentially silences larger intronless transgene constructs⁷, we hypothesized that the larger Cas9-BFP would be preferentially silenced, while the smaller BFP or BSD genes would not be silenced to the same degree. Cells were sorted for BFP to purity, and transgene expression was monitored over time by flow cytometry of BFP fluorescence (Fig. 1a, Methods). Indeed as we predicted, we observed significantly more silencing of Cas9-BFP ($P < 0.0001$; two-sided unpaired *t*-test) compared to the BFP transgene in both K562 and HEK293T cells over the course of a month (Fig. 1c).

To test whether the addition of an intron reduces silencing of Cas9, we inserted the second intron of the human β -globin gene (*HBB* IVS2) (850 bp), which has been reported to reduce transgene silencing⁷, into either the 5' untranslated region (UTR) or into the Cas9 gene body (Fig. 1b). The addition of the intron in either position significantly reduced the proportion of cells silencing Cas9-BFP in both cell lines ($P < 0.0009$; two-sided unpaired *t*-test) (Fig. 1c). To confirm silencing at the transcriptional level, we performed qPCR on K562 cells that had previously integrated Cas9 but had subsequently silenced it (BFP negative but blasticidin resistant), as well as cells still expressing Cas9-BFP (Supplementary Fig. 1A, Methods). As expected, we saw significantly reduced expression of Cas9 in the BFP-negative cells compared to the cells that had not silenced Cas9-BFP ($P < 0.0001$; two-way ANOVA test). However, we observed almost equivalent expression

of the shorter BSD transgene in the Cas9-silenced cells ($P > 0.1143$; two-way ANOVA test), suggesting that there was selective silencing of the longer Cas9-BFP transgene (Supplementary Fig. 1B). In addition, there was an increase in transcript expression of Cas9-BFP in the intron-containing cells compared to cells with intronless Cas9-BFP ($P = 0.0022$; two-way ANOVA test) (Supplementary Fig. 1C).

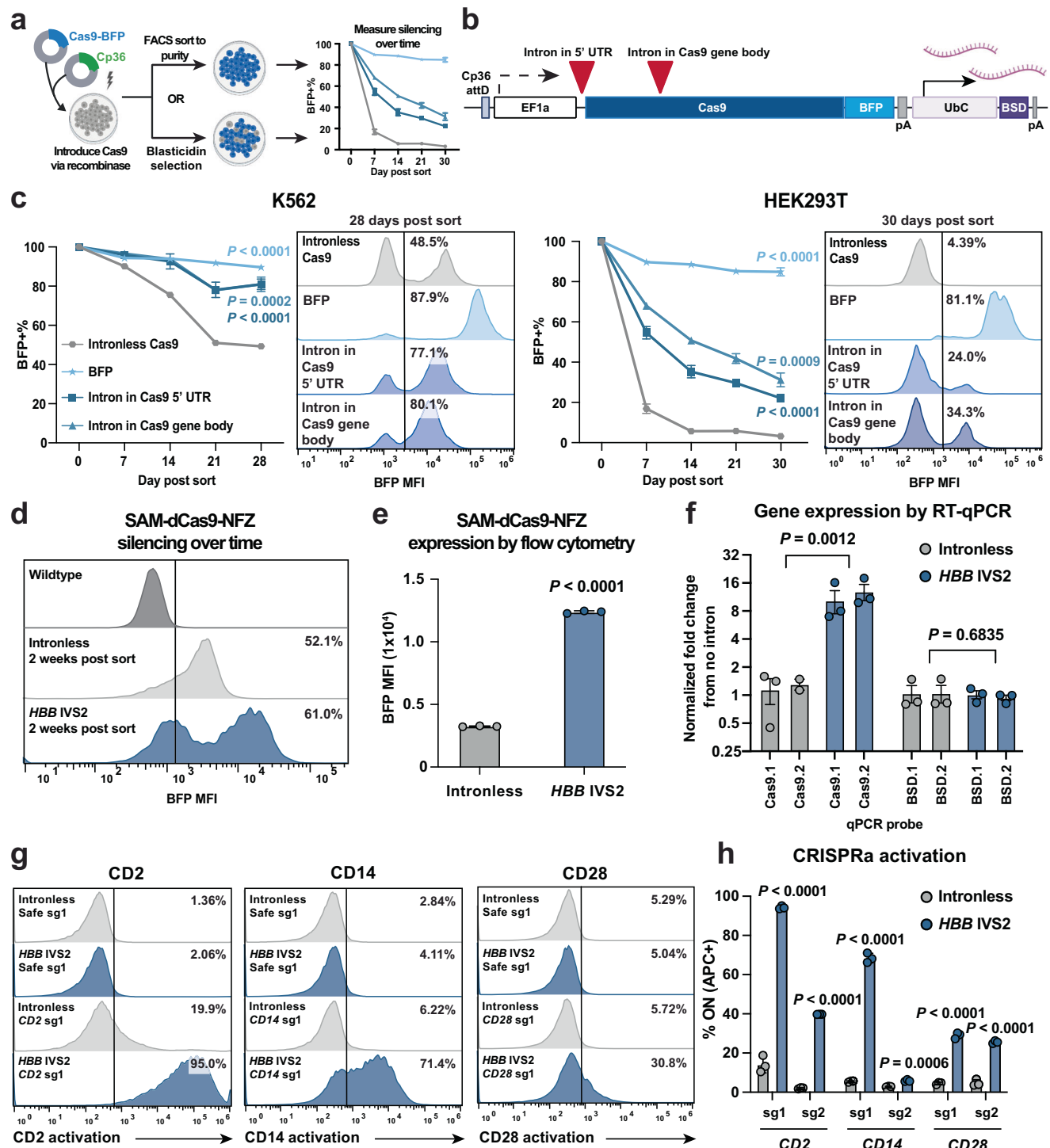
Because silencing can be highly variable across different cell and tissue types, we tested these constructs in human induced pluripotent stem cells (hiPSC) as well as murine lung adenocarcinoma LLC cells²⁷. As before, Cas9 was integrated via Cp36, and cells were selected for successful integration using blasticidin. In both cell lines, the addition of the *HBB* IVS2 intron significantly reduced the proportion of cells silencing Cas9-BFP ($P < 0.0387$; two-sided unpaired *t*-test) (Supplementary Fig. 2A), thus demonstrating that the addition of an intron may mitigate silencing across diverse genetic backgrounds.

Finally, we tested if insertion of the intron impaired Cas9 activity in K562, perhaps because of alterations to splicing, which might be problematic independent of their effect on silencing (Methods). Inserting the intron into the 5' UTR or the gene body did not impair Cas9 targeted endonuclease activity to a randomly inserted GFP reporter locus in cells expressing Cas9, suggesting that inclusion of the intron did not alter Cas9 activity (Supplementary Fig. 3A). Thus, the addition of the *HBB* IVS2 intron reduced Cas9-BFP silencing while maintaining transgene function in multiple cell types and species.

HBB IVS2 increases dCas9-activator expression and CRISPRa-mediated activation of target endogenous loci

Because we observed that addition of an intron increases transcript expression (Supplementary Fig. 1C), we hypothesized that the addition of an intron could improve the efficacy of CRISPR applications that require constitutive and robust expression to function, such as CRISPRa or CRISPRi, by increasing the abundance of dCas9-effector complexes. To test this, we cloned *HBB* IVS2 into a synergistic activation mediator (SAM) CRISPRa construct consisting of a tricistronic 2A-linkage of the PPH SAM domain, dCas9 fused to an NFZ activation domain, and BFP (hereafter referred to as SAM-dCas9-NFZ) (Supplementary Fig. 4A)^{28,29}. We then installed the construct into HEK293T cells using Cp36 recombination and sorted the cells by BFP expression to purity. Similar to what we observed with nuclease Cas9, the intron-containing CRISPRa construct was less transcriptionally silenced than the intronless CRISPRa construct (Fig. 1d). In addition, *HBB* IVS2 significantly increased transcript abundance of SAM-dCas9-NFZ by 11.6-fold over the intronless construct ($P < 0.0392$; two-sided unpaired *t*-test) (Fig. 1e, f). Consistent with the transcriptional measurements, we observed significantly higher BFP mean fluorescent intensity (MFI) of SAM-dCas9-NFZ by flow cytometry from the intron-containing compared to the intronless construct ($P < 0.0001$; two-sided unpaired *t*-test) (Fig. 1d).

Next, we infected the HEK293T lines stably expressing the SAM-dCas9-NFZ constructs with sgRNAs targeting *CD2*, *CD14*, *CD28*, or a safe-targeting intergenic locus³⁰ and measured the CRISPRa-mediated activation of the respective endogenous loci using flow cytometry of cell surface protein staining (Fig. 1g, h). In all instances, addition of the intron significantly enhanced activation of the target gene. By including *HBB* IVS2 in the CRISPRa construct, we increased the proportion of cells expressing CD2 by 4.7 fold and 15.4 fold (*CD2*-sg1 and *CD2*-sg2, respectively; $P < 0.0001$, two-sided unpaired *t*-test), cells expressing CD14 by 14-fold and 7.6-fold increase in activation by the intron-containing dCas9 (*CD14*-sg1 and *CD14*-sg2, respectively; $P < 0.00061$; two-sided unpaired *t*-test), and cells expressing CD28 by 12.8-fold and 7.7-fold ($P < 0.0001$; two-sided unpaired *t*-test). Importantly, sgRNAs that were ineffective with the intronless dCas9-effector (such as *CD28*-sg1 and *CD28*-sg2) now showed some measurable activity with the intron-containing effector, highlighting the limitation that low dCas9-effector expression can have on CRISPRa activity. These results show



that significantly more robust activation of endogenous loci can be achieved by using introns to increase the expression of CRISPRa constructs.

The *HBB* IVS2 intron protects against silencing irrespective of position in the Cas9 transgene

Having demonstrated the utility of introns in improving the expression of engineered transgenes, we were interested in elucidating the effect of different intron positions and sequences. We first examined if the effect of the *HBB* IVS2 intron is dependent on its position within the transcript, as 5' proximal introns have been reported to be enriched in gene regulatory activity^{18,31}. We cloned *HBB* IVS2 into 5 positions in the

Cas9-BFP transcript – in the 5' UTR, in the Cas9 coding sequence, between Cas9 and BFP, in the BFP coding sequence, or in the 3' UTR and integrated these constructs into K562 cells (Fig. 2a). Regardless of where the intron was placed in the transcript, it reduced Cas9-BFP silencing ($P < 0.0003$; two-sided unpaired *t*-test) (Fig. 2b). We observed a slight decrease in protective effect when the intron was placed in the 3' UTR of the transcript ($P = 0.0132$; two-sided unpaired *t*-test), although this construct still conferred significantly reduced silencing compared to the intronless transgene ($P = 0.0003$; two-sided unpaired *t*-test) (Fig. 2b). Consistent with previous reports that the intron must be within the transcriptional unit to enhance gene expression²², the intron did not mitigate silencing when placed

Fig. 1 | Impact of *HBB* IVS2 Intron Addition on Cas9 Transgene Silencing and CRISPRa-Mediated Activation. **a** Experimental setup for assessing transgene silencing. Cas9-BFP or BFP alone constructs were randomly integrated into K562 or HEK293T cells using the large serine recombinase Cp36. Flow cytometry analysis of BFP fluorescence was used to monitor transgene expression after either blasticidin selection or sorting to purity. Created in BioRender. Enright, N. (2025) <https://BioRender.com/s6u8bns>. **b** Vector cassette design for assessing transgene silencing. Red arrows represent locations where the *HBB* IVS2 intron was inserted. Created in BioRender. Enright, N. (2025) <https://BioRender.com/84hf4q7>. **c** Percentage of intronless Cas9-BFP, intron-containing Cas9-BFP, or BFP alone in K562 (left) and HEK293T (right) cells measured by flow cytometry up to one month after sorting. Data are mean of $n = 3$ biological replicates $\pm$ s.e.m.; two-sided unpaired t -test versus corresponding no-intron sample. Histograms show one representative biological replicate at the last measured time point. **d** Representative flow cytometry of silencing of intronless or intron-containing

SAM-dCas9-NFZ HEK293T cells as measured by percentage of BFP+ cells. Cells were measured two weeks after sorting. **e** Mean fluorescent intensity (MFI) of intronless or intron-containing SAM-dCas9-NFZ (BFP+) containing HEK293T cells. Data are means of $n = 3$ biological replicates $\pm$ s.e.m.; two-sided unpaired t -test versus corresponding no-intron sample. **f** Quantification of dCas9-NFZ and BSD mRNA abundance using RT-qPCR in intronless or intron-containing SAM-dCas9-NFZ HEK293T cells. Data are means of $n = 3$ biological replicates $\pm$ s.e.m.; two-way ANOVA test. **g** Representative histograms of CRISPRa-mediated activation in HEK293T cells at endogenous CD2, CD14, or CD28 loci with either intronless or *HBB* IVS2 intron-containing SAM-dCas9-NFZ. **h** The percentage of cells ON (APC) after gating for the sgRNA (mCherry+) and intronless or intron-containing SAM-dCas9-NFZ (BFP+). sgRNAs targeting CD2, CD14, or CD28 were first installed by lentiviral delivery into HEK293Ts and then selected by puromycin. Data are means of $n = 3$ biological replicates $\pm$ s.e.m.; two-sided unpaired t -test versus corresponding no-intron sample. Source data are provided as a Source Data file.

upstream of the promoter (Supplementary Fig. 5A, B). Analysis of BFP MFI showed higher protein expression of Cas9-BFP when *HBB* IVS2 was in the coding region between Cas9 and BFP ($P = 0.0012$; two-sided unpaired t -test), lower expression when it was in the 3' UTR ($P = 0.0109$; two-sided unpaired t -test), and no significant changes when placed in the other positions ($P > 0.1426$; two-sided unpaired t -test) (Supplementary Fig. 5C).

Increased intron length is correlated with increased protection from silencing

Previous results have suggested that *HBB* IVS2 is not unique in its ability to reduce transgene silencing and that other intron sequences may have similar effects⁷. However, it is unclear what intron properties govern their effects on silencing. We selected a small panel of introns spanning various sequence properties, such as length, T content, and structuredness, (Supplementary Data 1) and inserted them into the 5' UTR of the Cas9-BFP transcript to measure their effect on silencing³² (Fig. 2c). Introns were selected from essential genes in order to enrich for transcripts that are actively processed by the cell. We tested these in K562, again selecting for cells containing a successful integration and measuring silencing of the transgene over time after sorting to purity (Fig. 2d). We observed a significant correlation between intron length and the ability to reduce silencing ($R^2 = 0.573$; $P = 0.00275$, two-sided F-test; Supplementary Fig. 6A). Introns over 2 kb all performed better than the 850 bp *HBB* IVS2, while those similar in length performed similarly, and those shorter than *HBB* IVS2 performed worse. Further analysis revealed that those which performed well tended to be more T-rich and GC-poor (Supplementary Fig. 6B, C), consistent with previous studies^{7,33}. We analyzed other intron sequence properties, stratifying by intron length, and also found a positive association with the number of RNA-binding protein motifs ($P = 0.0025$, two-sided F-test, Supplementary Fig. 6D). Evolutionary conservation and structuredness of introns did not predict their ability to reduce silencing amongst the sequences we tested (Supplementary Fig. 6E–G).

To confirm these findings, we selected a subset of introns to validate in HEK293T cells, a cell line that robustly silences Cas9 expression (Fig. 1c, d). Concordant with what we observed in K562 cells, *HBB* IVS2, *HNRNPK* IVS3, and *PSMD14* IVS2 all protected against silencing of Cas9 ($P < 0.0001$; two-sided unpaired t -test), while *EEF2* IVS9 and *PGK1* IVS3 did not (Fig. 2e, f). Similar to what we observed in K562, *PSMD14* IVS2 was significantly better at reducing silencing than *HBB* IVS2 (Fig. 2e; $P < 0.0001$; two-sided unpaired t -test), while also maintaining high expression of Cas9-BFP (Fig. 2f). Despite being less than half the length of *HBB* IVS2, *HNRNPK* IVS3 was also significantly better at reducing silencing in HEK293T cells (Fig. 2e, f; $P < 0.0001$; two-sided unpaired t -test). Examination of the sequence properties of these two introns revealed that they both had higher thymine content and more predicted RBP binding motifs than the other introns of similar length that we tested (Supplementary Fig. 6A). Our results

indicate that diverse introns are capable of reducing silencing of Cas9, and the protective effect of an intron is correlated with its length and thymine content.

Combining introns with a UCOE provides synergistic protective effect against silencing

Since introns are expected to reduce silencing through escaped RNA recognition from the HUSH complex⁷, we hypothesized that DNA sequence elements working in orthogonal ways could synergistically reduce silencing. To find elements that could further improve protection from silencing, we introduced active or passive DNA regulatory elements to our construct: either a minimal universal chromatin opening element from the A2UCOE or flanking core *CHS4* insulator elements^{34–36} (Fig. 2g). Ubiquitous chromatin opening elements (UCOE) are thought to promote open chromatin through bidirectional promoter activity and CpG-rich sequences, while insulator sequences are thought to reduce silencing by preventing the spread of heterochromatin^{35,37,38}. We found that, compared to *HBB* IVS2, the *CHS4* insulator was ineffective at reducing silencing of Cas9-BFP in K562, and the combination of *CHS4* insulators and *HBB* IVS2 performed similarly to *HBB* IVS2 alone (Fig. 2h). The addition of the UCOE alone was able to reduce silencing ($P = 0.0018$; two-sided unpaired t -test), at a level that was comparable to that of *HBB* IVS2 alone (Fig. 2h). Combination of the UCOE with *HBB* IVS2 displayed a synergistic protective effect (Fig. 2h; $P = 0.0005$; two-sided unpaired t -test). Validation of these UCOE constructs in HEK293T cells showed that the UCOE alone was better at reducing silencing than *HBB* IVS2 (Fig. 2i; $P = 0.0157$; two-sided unpaired t -test) but did not have the effect of increasing gene expression, as assessed by BFP MFI (Fig. 2j). Strikingly, the combination of both elements was significantly more effective at reducing silencing in HEK293T cells than either element alone, with the additional benefit of increasing Cas9-BFP expression (Fig. 2i, j; $P < 0.0001$; two-sided unpaired t -test). To assess whether this effect could be reproduced with another transgene, we tested the addition of *HBB* IVS2 and UCOE, either alone or in combination, with the Cas12a-BFP transgene (Supplementary Fig. 7A). After integrating the constructs into HEK293T cells and selecting with blasticidin, we again observed that *HBB* IVS2 significantly reduced silencing compared to the intronless construct ($P < 0.0001$; two-sided unpaired t -test) and that the combination of *HBB* IVS2 with UCOE further reduced silencing compared with either element alone (Supplementary Fig. 7B, C; $P < 0.0001$; two-sided unpaired t -test). These results suggest that combining DNA-level regulators with introns, which work at the RNA level, can synergistically reduce the silencing of Cas9 or Cas12a.

Discussion

In this study, we explored the potential of introns to prevent Cas9 silencing and increase expression^{7,20}, which is a pervasive technical barrier in gene editing experiments³. We identified several

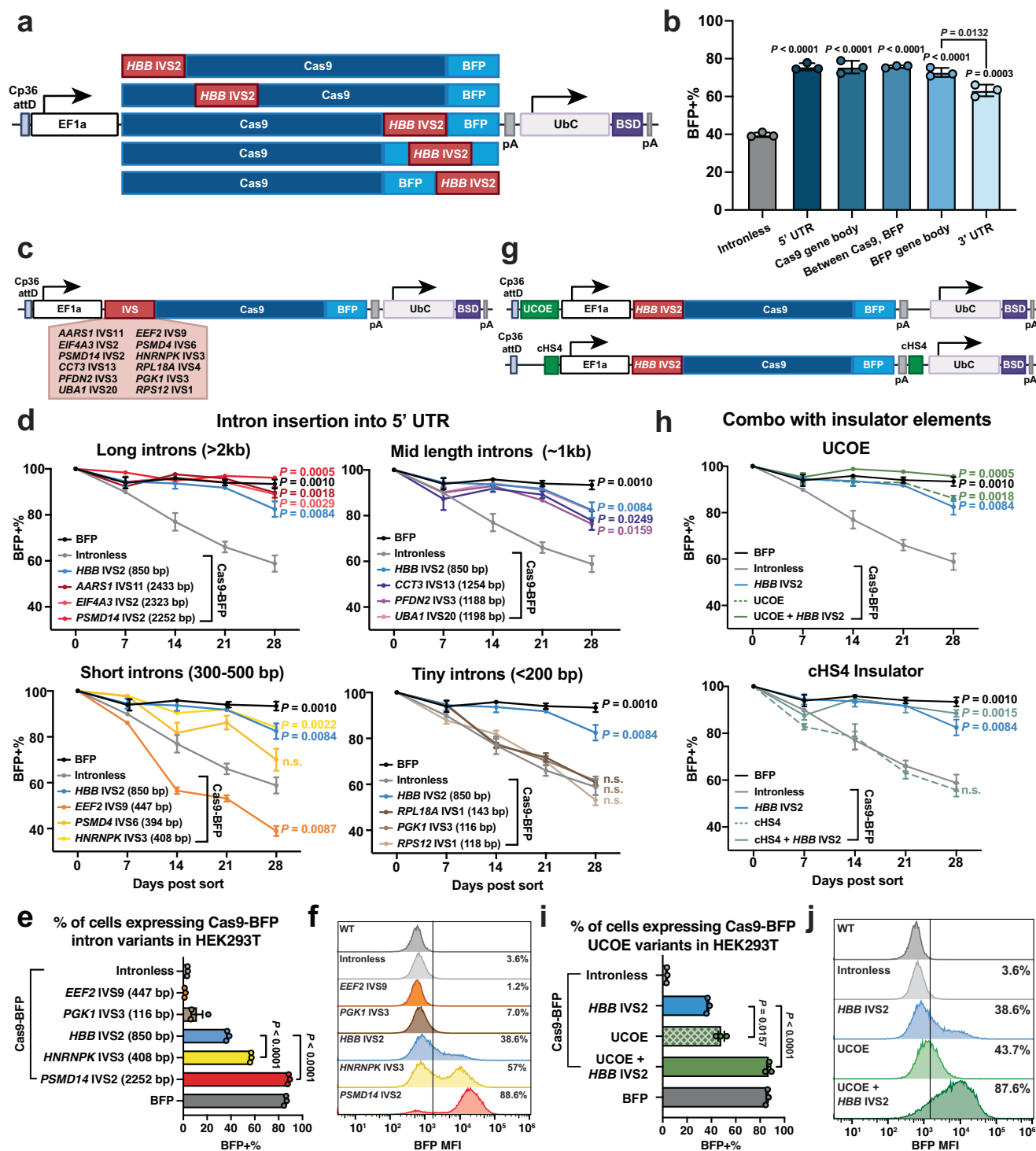


Fig. 2 | Anti-silencing effects of introns and synergistic effects with DNA regulatory elements. **a** HBB IVS2 was cloned into five different positions in the Cas9-BFP transcript: in the 5' UTR, Cas9 gene body, between Cas9 and BFP, BFP gene body, or the 3' UTR. **b** Percentage of Cas9-BFP expressing cells in K562 cells with intronless or intron-containing Cas9-BFP measured by flow cytometry after blasticidin selection. Data are means of $n = 3$ biological replicates $\pm$ s.e.m.; two-sided unpaired t -test versus corresponding no-intron sample and versus intron in 3' UTR. **c** A panel of introns from essential genes of varying lengths and sequence characteristics were inserted into the 5' UTR of the Cas9-BFP transcript. **d** K562 cells were monitored for transgene expression by flow cytometry for one month after sorting to purity. Data are means of $n = 3$ biological replicates $\pm$ s.e.m.; two-sided versus corresponding no-intron sample, unpaired t -test at the final time point. **e** Validation of select introns from (d) in HEK293T cells after selection of cells with blasticidin. Data are means of

$n = 3$ biological replicates $\pm$ s.e.m.; two-sided unpaired t -test. **f** Representative BFP MFI from flow cytometry of select introns variants from (d) in HEK293T cells after blasticidin selection. **g** A minimal universal chromatin opening element (A2UCOE) was cloned upstream of the TSS of both intronless and intron-containing Cas9-BFP and two cHS4 insulators were cloned to flank intronless and intron-containing Cas9-BFP. **h** K562 cells were monitored for transgene expression by flow cytometry for one month after sorting to purity. Data are means of $n = 3$ biological replicates $\pm$ s.e.m.; two-sided versus corresponding no-intron sample, unpaired t -test at the final time point. **i** Validation of combination of HBB IVS2 with UCOE from (h) in HEK293T cells after selection of cells with blasticidin. Data are means of $n = 3$ biological replicates $\pm$ s.e.m.; two-sided unpaired t -test. **j** Representative BFP MFI from flow cytometry of UCOE variants from (h) in HEK293T cells after blasticidin selection. Source data are provided as a Source Data file.

properties of introns that contribute to their effectiveness in mitigating silencing and demonstrate that this can markedly enhance Cas9 expression across a range of cell types. Inserting the *HBB* IVS2 intron into multiple positions in the Cas9 transcript revealed that placing the intron in any position was sufficient to significantly reduce silencing compared to the base construct. Testing a larger panel of introns with different sequence characteristics, we observed a strong correlation between intron length and the ability to prevent silencing. Nuclease Cas9 constructs containing an intron were markedly less prone to silencing and retained high endonuclease activity, while CRISPRa constructs with an intron also maintained expression and were more potent in activation of target gene expression. These findings suggest the broad utility of including introns in transgene constructs to reduce Cas9 silencing, which should allow for more stable transgene expression and efficient gene editing.

In addition to various intron sequences, we tested the synergistic potential of combining introns with insulators including A2UCOE and cHS4. The cHS4 insulator alone did not reduce silencing of Cas9, and when used in combination with *HBB* IVS2, it did not appear to provide additional benefit against silencing. In contrast, the addition of a UCOE by itself was able to reduce silencing. Whereas cHS4 has been proposed to act as a passive boundary element that blocks the spread of heterochromatin³⁴, UCOE sequences are known to promote active transcription and open chromatin^{35–37}. The greatest reduction of silencing was observed when using the UCOE in combination with *HBB* IVS2. The enhanced protection against silencing observed with the UCOE in conjunction with the *HBB* IVS2 intron suggests that regulatory elements acting at the DNA level can complement RNA-based mechanisms of protection to further reduce silencing.

We observed that introns not only increased the proportion of cells expressing the transgene but also enhanced the overall expression of the transgenes. We consistently found increased transcript levels of intron-containing Cas9 which corresponded to an increase in Cas9 protein. This may be especially beneficial for epigenome editing applications such as in CRISPRa or CRISPRi technologies³⁹, where low effector protein expression has been shown to be a challenge for robust and sustained epigenome editing. Here, we found that including *HBB* IVS2 in a CRISPRa construct resulted in more effective activation of endogenous target genes, as evidenced by increased expression levels of CD2, CD14, and CD28 in HEK293T cells, whereas the intronless versions were unable to activate these target genes.

We noticed different degrees of silencing in different cell lines: in particular, HEK293T and hiPSC silenced Cas9-BFP much more potently than K562 cells. This may be expected, as mechanisms and strength of silencing are known to be variable across different cell types³. Previous work has reported HDAC4-mediated silencing to be faster in HEK293T cells than K562 cells⁴⁰, and the silencing of short sequences such as BFP in hiPSC cells suggests that mechanisms other than HUSH may contribute to silencing in these cells^{41,42}. Other cell-line specific effects may determine the efficiency with which intron inclusion prevents silencing. For example, splicing efficiency may differ across cell types. Although this may not always affect the protective effects of the intron against silencing, this may affect transgene function if splicing is incomplete and occurs within the coding region. Placing an intron in the 5' UTR could minimize this effect. Future work will be required to fully investigate diverse mechanisms of silencing across different cell types, and optimize how the inclusion of introns and other anti-silencing elements may improve transgene expression.

Adding intron sequences increases the size of gene expression constructs, which is often a limitation for many modes of nucleic acid delivery, such as viral packaging and electroporation. Although we observed that the protective effect of an intron is significantly correlated with intron length (consistent with previous observations⁷), we discovered a small intron *HNRNPK* IVS3 (408 bp) that was as effective

as the longer *HBB* IVS2 (850 bp) at mitigating silencing. In our analysis of intron sequence properties, we found that introns with higher T content and more predicted RNA-binding protein sites tended to reduce silencing better within introns of similar length. These principles could be applied to find or engineer short introns that are effective at reducing silencing. While we did not find strong associations with intron secondary structure, we only examined a small selection of introns in this study, and future studies could screen larger panels of introns to elucidate other intron sequence and structural properties that affect silencing.

In this study we avoided potential confounding effects of lentiviral delivery by integrating expression cassettes using the large serine recombinase Cp36, which recombines large DNA sequences directly into the genome with the addition of an attachment site. Although lentiviral vectors are commonly used for transgene expression, introns in lentiviral vectors can be spliced out at the RNA stage of the viral life cycle and thus not incorporated into the genome. Other studies have worked around this issue by cloning the internal cassette in the anti-sense orientation relative to the viral genome⁷. This same study found that introns with splice-site mutations also protected against HUSH repression, a strategy which could allow for lentivirus to be made without cloning the cassette in the anti-sense orientation.

Although we focused here on maintaining Cas9 expression, our results suggest that the inclusion of introns in expression cassettes will likely be useful for any synthetic biology application that requires robust expression of longer transgenes. Future studies will more completely define the mechanisms by which introns prevent silencing, as well as the complementary regulatory elements that can combine to ensure stable expression across a range of diverse cell types and chromatin environments.

Methods

Cancer cell culture

HEK293T (CRL-3216), K562 (CCL-243), and LLC (CRL-1642) cells were obtained from ATCC. HEK293T and LLC cells were grown in DMEM (Gibco 11995073) supplemented with 10% heat-inactivated FBS (Cytiva SH30910.03), penicillin/streptomycin (100 U mL⁻¹) (Gibco 15140163), and GlutaMAX (100 U mL⁻¹) (Gibco 35050061) and passaged with TriPLE (Thermo Scientific 12605028) or 0.25% trypsin-EDTA (Thermo Scientific 25-200-114). K562 cell lines were maintained in suspension culture in RPMI-1640 (Gibco 11875119) supplemented with 10% FBS, penicillin/streptomycin (100 U mL⁻¹), and GlutaMAX (100 U mL⁻¹). All cells were cultured in a humidified 37 °C incubator set at 5% CO₂ and passaged 2–3 times weekly.

hiPSC culture

WTC-11 (UCSF1001-A) hiPSC cells were obtained from the Allen Institute for Cell Science. The hiPSC lines were cultured in feeder-free, serum-free mTeSR Plus medium (STEMCELL Technologies 100-0276). All cells were cultured in a humidified 37 °C incubator set at 5% CO₂ and passaged 2–3 times weekly by lifting with Accutase (Innovative Cell Technologies Inc AT104) and plating onto tissue culture dishes previously coated for 1 h at 37 °C with Geltrex (Gibco A1413302) diluted in DMEM/F-12 (Gibco 11320033), in mTeSR Plus medium supplemented with 10 μM Y-27632 dihydrochloride (Medchem Express HY-10583) for the first 24 h post-passaging. Blasticidin selection was carried out at 10 μg/mL (Thermo Scientific A1113903).

Cloning

All Cas9, dCas9, and Cas12a constructs were cloned using the Gibson Assembly method as described previously⁴³, using NEBuilder HiFi DNA Assembly Master Mix (New England Biolabs E2621L).

The GFP-targeting sgRNA (5'-GGCGAGGGCGATGCCACCTA-3') was cloned into a GFP-puro backbone (pMCB306). The plasmid was linearized using a double BstXI and BlnI digest (BstXI: New England

Biolabs R0113S, BpI: New England Biolabs R0585S) and the guide oligo was ligated using T4 DNA ligase (New England Biolabs M0202S).

The *HBB* IVS2 intron was PCR amplified from the pHRISIN-as(PSFV-GFP*(-1)-ATGless-ORF2-(5'-HBB IVS2)-SV40pA)-pSV40-Blast plasmid (gift from P. Lehner)⁷. Other introns were either PCR amplified out of KOLF hiPSC (*AARS1* IVS11, *EIF4A3* IVS2, *CCT3* IVS13, *PFDN2* IVS3) or MCF7 (*PSMD14* IVS2) genomic DNA using CloneAmp HiFi PCR Premix (Takara Bio 639298) or synthesized (all others) (Twist Biosciences) and cloned into backbone plasmids using Gibson Assembly.

Plasmids in this manuscript will be made available on Addgene and are listed in Supplementary Data 2.

Flow cytometry

Live cells were analyzed on an Attune NxT Flow Cytometer (Thermo Fisher Scientific). Cell sorting was carried out on a FACSaria III (BD Biosciences). Sorted cells to be maintained in culture were kept on ice in MACS buffer (4 mM EDTA 2% heat-inactivated FBS in PBS (Gibco 10010023) during sorting, and sorted into FBS (Cytiva SH30910.03) with penicillin/streptomycin (100 U mL⁻¹) (Gibco 15140163).

Data were analyzed using CellEngine (CellCarta, Montreal, Canada) software and visualized using FlowJo v10.8.1 Software (BD Life Sciences).

Transgene insertion

Cp36 was used to randomly integrate donor DNA into the genome at pseudosites without the need to engineer attachment sites²⁶. To create all Cas9 K562 cell lines, 1.0×10^6 K562 cells were electroporated in 100 μ L of Amaxa solution (Amaya I: ATP Disodium Salt: 0.2 M, MgCl₂ · 6H₂O: 0.12 M; Amaya II: KHPO₄: 12 mM, NaHCO₃: 1.2 mM, Glucose: 0.4 mM, pH 7.4) using a Lonza Nucleofector 2b (program T-016). Approximately 2 μ g of an equimolar mixture of the Cas9 donor plasmids and the 5.7-kb Cp36 plasmid were used.

To create all Cas9, SAM-dCas9-NFZ, and Cas12a HEK293T or LLC cell lines, ~2 μ g of donor plasmid and Cp36 plasmid were co-transfected at equimolar amounts using Lipofectamine 3000 following the manufacturer's recommendations (ThermoFisher L3000001). Cells were seeded to be 70% confluent at time of transfection.

To create WTC-11 hiPSC cell lines, 8.0×10^5 cells were electroporated in 100 μ L Human Stem Cell Nucleofector solution 1 (Lonza VPH-5012) using a Lonza Nucleofector 2b (program B-016), and plated into one well of a 6-well plate. Approximately 2 μ g of donor plasmid and Cp36 plasmid were electroporated at equimolar amounts. Medium was supplemented with 10 μ M Y-27632 dihydrochloride (Medchem Express HY-10583) one hour before cells were lifted for nucleofection and for 24 h after nucleofection.

All experiments were performed with at least three biological replicates, resulting from independent transfection events.

Cas9 efficiency test

To assess Cas9 efficiency, we infected the WT or Cas9-expressing K562 cells with lentivirus containing GFP-targeting sgRNA in a GFP-puro backbone (pMCB406) at an MOI of <0.3. 48 h after transduction, cells began selection with puromycin (Cayman Chemical 28240) at a concentration of 1 μ g/mL. Cas9 endonuclease activity (GFP depletion) was measured using an Attune NxT flow cytometer after 2 weeks in Cas9-BFP expressing cells.

Bioinformatic analysis

Candidate genes for intron selection were picked from essential genes⁴⁴. Introns from essential genes were filtered for splicing at >97% efficiency in K562, as determined using ENCODE RNAseq data, with splicing efficiency estimated using SPLICE-q^{45,46}. All introns from candidate genes were extracted according to GENCODE v29 annotations from the GRCh38 reference genome. Three introns from four length bins (~2 kb, ~1 kb, 300–500 bp, <200 bp) were selected that reflected a

range in T content and predicted free energies, calculated by ViennaRNA⁴⁷. Mouse and human RBP binding motifs were detected using RBPmap v1.2⁴⁸. Conservation scores from phastCons and phyloP were accessed from UCSC genome browser for the hg38 genome and the average score was taken over each intron locus^{49,50}. Association testing between silencing and number of RBP binding motifs and predicted minimum free energy was performed using a random mixed effects model using the nlme package (v3.1-163) with length as a random variable. Association testing between silencing and all other intron sequence properties was performed using a two-sided F-test in R (v4.2.1).

Intron position and sequence testing

After transgene insertion, cells were allowed to recover for 72 h before selection with 10 μ g/mL blasticidin. While selecting, cells were split and analyzed by flow cytometry using an Attune NxT every 2–3 days.

To measure silencing over time, cells were sorted to purity using fluorescence-activated cell sorting (FACS) (FACSaria III, BD Biosciences) so that the BFP+ population was 100%. Cells were split every 2–3 days and analyzed by flow cytometry once a week for 4 weeks using an Attune NxT. The cells were not kept on blasticidin during this time.

Lentiviral production

To produce lentivirus containing the sgRNAs used for CRISPRa and the Cas9 efficiency test, HEK293T cells were plated as to be 70–80% confluent at the time of transfection. Each well was transfected with 750 ng of an equal mass mix of the third-generation packaging plasmids (pMD2.G Addgene #11259, psPAX2 Addgene #12260, and pMDLg/pRRE Addgene #12251), 750 ng of sgRNA plasmid, and 10 μ g polyethylenimine (PEI) (Fisher Scientific NC1014320) mixed in 200 μ L serum-free DMEM (Gibco 11995073). The mixed transfection reagents were incubated at room temperature for 20 min and then added dropwise to the HEK293T cells. Lentivirus was harvested at 48 h and 72 h after transfection and passed through 0.45 μ m PVDF filtration units (EMD Millipore SLHV033RB) before storage at -80 °C until used.

RT-qPCR

For all RT-qPCR, 250k cells were sorted and washed with PBS (Gibco 10010023) before being lysed in 400 μ L Tri Reagent and extracted according to manufacturer's recommendations (Zymo Research, R2061). High-Capacity RNA-to-cDNA (Applied Biosystems 4387406) was used to generate cDNA, according to manufacturer recommendations. qPCR were set up with three technical replicates for each sample with human Taqman gene expression probes (Taqman Probe ID: Hs02758991.g1 FAM dye for *GAPDH*, Hs00801390.s1 VIC dye for *TUBB3*). Integrated DNA Technologies: Cas9 FAM Probe 1: FWD: GAAACGAAGTGGCCCTGCCCTC, REV: GGGGGAGCCCTTCAGCTTCTCA, Probe: TTCCTGTACCTGGCCAGCCACT; Cas9 FAM Probe 2: FWD: ACGAGATGGCCAAGGTGGACGA, REV: CCACCTCGTCCACGATGTTGCC, Probe: TAAGAAGCACGAGCGGCACCCC; BSD FAM Probe 1: FWD: GAAGACTACAGCGTCGCCAGCG, REV: ACAAGTCAGGTTGCCAGCTGCC, Probe: GCAGAAGTCTGGTGCTGGGCA; BSD FAM Probe 2: FWD: GAAGACTACAGCGTCGCCAGCG, REV: CCCAGCACCACGAGTTCTGCAC, Probe: CGACGGCCGCATCTTCACTGCT. Cas9 and BSD primers were designed using Primer3. All qPCR reactions were run on a Bio-Rad CFX384 using Taqman Fast Advanced Master Mix (Applied Biosystems™ 4444556). Ct values normalized to *TUBB3* and *GAPDH* housekeeping genes.

CRISPRa

HEK293T cells expressing intronless or intron-containing PPH-T2A-dCas9-NFZ-P2A-BFP were infected with lentivirus expressing an sgRNA at <0.3 MOI. sgRNAs used were the following: *CD2*-sg1: 5'-GCAAAG-GAGCACATCAGAA-3'; *CD2*-sg2: 5'-GCACGTGGTTAAGCTCTCG-3'; *CD14*-

sg1: 5'-GTAACAGGAAGGATTCTGCA-3'; *CD14*-sg2: 5'-GCTCTGGGATA-TAAGGTAGG-3'; *CD28*-sg1: 5'-CTGCAGTCAGGATGCCTTG-3'; *CD28*-sg2: 5'-GATGTGCCCTAAGGGGATGG-3'; Safe-sg1: 5'-GAGGAGACCAATGA TCTCT-3'. Cells were selected to purity using puromycin and confirmed via mCherry fluorescence. Cell surface staining for activation of CD2, CD14, and CD28 was performed with APC-conjugated antibodies (CD2: Miltenyi Biotec 130-116-150 at 1:20, CD14: Biolegend 367117 at 1:100, CD28: Invitrogen 17-0289-42 at 1:100) and proportion and intensity of gene activation was measured using an Attune NxT flow cytometer.

Statistics and reproducibility

Statistical details are reported in the figures and figure legends. Statistics were calculated using GraphPad Prism (v10.3.0) or in R (v4.2.1). All figure panels show representative data from at least three independent biological replicates. No statistical method was used to pre-determine sample size. RT-qPCR technical replicates with normalized Ct values that deviated by >1 cycle were excluded from analysis. The experiments were not randomized. The Investigators were not blinded to allocation during experiments and outcome assessment.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this study are available within the paper and its Supplementary Information. Source data are provided with this paper.

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Author contributions

S.A. and P.P.D. designed, performed and analyzed the experiments. S.A., P.P.D. and N.E. cloned the Cas9 and Cas12a plasmids. K.S. cloned the plasmid and generated lentivirus for the Cas9 endonuclease activity test. S.A., A.V.J. and A.V. performed the hiPSC culture. P.P.D. performed all bioinformatics analyses. R.V., C.A.G., T.N. and A.V. assisted with cell culture and collection and interpretation of flow cytometry data. L.M.P., P.J.L., M.C.B. and S.A. acquired funding. L.B., M.S., and P.J.L. assisted with experimental design and interpretation. S.A., P.P.D. and M.C.B. conceived the study and wrote the manuscript with input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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